Dulari Bhushan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
Tagore Public School	Class XII (CBSE)	2021	87.00%
Tagore Public School	Class X (CBSE)	2019	95.60%

PROJECTS

REFUTE: Falsification-First KPI Intelligence [Link] | Accenture Hackathon

- **Built** a KPI root-cause engine that tries to *disprove* each explanation, giving every cause a KILLED, SURVIVED or INCONCLUSIVE verdict. An LLM (Qwen, OpenRouter or Bedrock) only proposes tests, and statistical models (Bayesian changepoint, difference-in-differences) decide the outcome.
- **Delivered** tailored briefs for Leaders, Managers and Engineers from one verdict ledger, each with role-based access controls and SQL-traceable evidence.
- **Visualised** an interactive knowledge graph of KPIs, sources and hypotheses, showing what connects to what and which causes survived or were ruled out.
- **Deployed** the engine live on AWS (Docker, ECS Fargate), with Bedrock behind a scoped IAM role and validated outputs.

Continual Domain Generalization with Concept Bottleneck Models | *M.Tech Thesis Advisor: Dr. Srijith P.K.*

- **Reproduced** the LanCE continual domain generalization baseline, fixing five implementation bugs across its DDO regularization and CLIP-based concept embeddings, to establish a validated experimental foundation for the thesis.
- **Compared** EWC and replay-based continual learning strategies across domain shifts, characterizing distinct forgetting patterns and exposing limitations of fixed concept vocabularies.
- **Designing** concept bottleneck mechanisms with sequentially adaptable concept spaces to sustain reliable, interpretable predictions as new visual domains emerge.

3D LiDAR Semantic Segmentation using RangeViT [Link] | *Faculty: C. Krishna Mohan*

- **Reproduced** and extended a Vision Transformer-based range-view pipeline on SemanticKITTI's sparse LiDAR scans, establishing a working baseline for 3D semantic segmentation.
- **Engineered** decoder-side **geometry fusion** to recover structural cues lost in 2D range-view projection, boosting boundary-level segmentation accuracy.
- **Designed** prototype-based feature supervision to enforce structured per-class representations, enabling **interpretability** without sacrificing segmentation performance.
- **Validated** the design through controlled ablations: prototype-only supervision reduced mIoU from **0.3629 to 0.27**, geometry-aware integration recovered performance to **0.3861**, and the full configuration reached **0.3922 mIoU.**

WORK EXPERIENCE

Firmware Engineer Intern | Volansys Technologies, ACL Digital

- **Automated** MAC address retrieval, verification and **device flashing** via Google Sheets APIs, with **real-time status tracking** that cut manual provisioning effort and errors.
- **Engineered** a contactless **camera-based test jig** for a manufacturing line that checks a **DUT's LED** across all RGB colours and runs the full sequence in different permutations, automatically flagging any broken or misbehaving LED.

AI Research Intern | CHARUSAT University

- **Developed** a transfer-learning ensemble using ResNet-34, ResNet-50, GoogleNet, VGG and ViT for seven-class emotion detection, achieving **86% accuracy**.
- **Applied SHAP** for fine-grained feature-importance analysis, uncovering model bias and improving transparency of predictions.

SKILLS AND CERTIFICATIONS

Programming: C++, C, Python, SQL

Technical: Machine Learning, Deep Learning, Computer Vision, Representation Learning, Continual Learning, Domain Generalisation, Generative AI (LLMs), Prompt Engineering, AI-Assisted Product Development

Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, Copilot SDK, Claude Code

Cloud & Tools: AWS, Microsoft Azure, Git/GitHub, Figma, Power BI

Certifications: Microsoft Azure AI Fundamentals | Google Advanced Data Analytics

RESEARCH PUBLICATIONS

DeViTC: Deep Vision Transformer for Currency Originality Recognition [Link]

IEEE Computer Journal, 2025

- **Built** a custom dataset of **10,000+ high-resolution RGB images**, applying preprocessing and augmentation pipelines to enable robust currency-recognition model training.
- **Benchmarked** four architectures: Vision Transformer, CNN, ResNet, and EfficientNet, through systematic hyperparameter optimisation, achieving state-of-the-art **99.56%** classification accuracy.
- **Evaluated** generalisation across Ghanaian, Ethiopian, Thai and Peruvian currency datasets to demonstrate robustness across visually diverse currency domains.

Ensemble Graph Attention Network with U-GAT for Alzheimer's Detection [Link]

International Journal of Technology (IJTech), 2025

- **Designed** an ensemble combining U-Net and Graph Attention Networks, processing **86,000+ OASIS MRI images**, to enable Alzheimer's detection from structural brain scans.
- Corrected **severe class imbalance** (67,200+ Non-Demented vs. under 500 Moderate) through targeted preprocessing and augmentation, achieving balanced recall across dementia stages.
- **Achieved 96% accuracy**, outperforming CNN, VGG19, EfficientNet+ViT, Multi-UNet and attention-based architectures, proving the strength of graph-based relational learning.

DResNet: Deep Residual Network for Handwritten Character Recognition [Link]

Conference on Computer Vision and Robotics (CVR), 2024

- **Fine-tuned** ResNet50 and benchmarked it against VGG and CNN architectures, optimising layers and hyperparameters to maximise handwritten-character recognition performance.
- **Tested** across **Gujarati, Urdu & Devanagari** handwritten datasets, achieving recognition accuracies of **99.71%, 99.67%** and **99.17%**, respectively.

Leveraging Explainable AI in Deep Learning for Brain Tumor Detection [Link]

Explainable AI in Healthcare Imaging for Medical Diagnoses (Book Chapter)

- **Synthesised** 71 sources across five deep learning paradigms (CNN, RNN/LSTM, Transformer, autoencoder, GAN) for MRI brain tumor detection, mapping strengths against **9 clinical deployment challenges**.
- **Assessed transfer learning & ensemble strategies** (bagging, boosting, voting on VGG16, ResNet, DenseNet backbones) for improving segmentation on **limited annotated MRI datasets**.
- **Evaluated** Grad-CAM, SHAP and attention-based interpretability methods against clinical trust and regulatory needs, positioning **explainability** as key to **safe AI** adoption in radiology.

RELEVANT COURSES

Advanced Data Structures and Algorithms, Convex Optimisation, Deep Learning, Foundations of Machine Learning, Matrix Theory, Probability and Stochastic Processes, Visual Computing, Data Science Analysis

POSITIONS OF RESPONSIBILITY

Placement Coordinator, IIT Hyderabad

- Coordinated placement communications and recruiter interactions across student batches, streamlining scheduling and reducing follow-up delays during the placement cycle.

Club Captain, AWS Cloud Clubs, CHARUSAT

Selected as one of 87 AWS Cloud Clubs Captains worldwide

Led AWS Student Community Day: secured a **$3,000 AWS grant**, coordinated **15 national & international experts**, and **led 50 volunteers** to deliver a technology event reaching **1,000+ attendees**.

Hosted **international & local GenAI experts** for **500+ students**, expanding exposure to applied generative AI use cases.

Designed and delivered hands-on AI-in-Cloud training, including a **5-week AWS program** culminating in projects built by 20+ participants.

Led women-focused technology initiatives: organized three Women in **AI** panels and workshops impacting 450+ students, and conducted a **Cloud** and **FinOps** session for 150+ students with industry professionals.

Delivered an **international session** on Edge Computing to **3,000+** attendees at **AWS User Group China**, covering modern edge architectures and distributed computing use cases.

Invited by **AWS Cloud Clubs FUTA, Nigeria**, to deliver an AI fundamentals session for 20+ students, covering real-world applications and pathways to becoming an AI engineer.

AI/ML Lead, Club Gamma by GitHub

- Led AI/ML sessions and project discussions for **150+ students** across two semesters, driving technical engagement and peer learning within the university community

Formality and Finance Manager, Team Environment CHARUSAT

- Coordinated sustainability and community initiatives including an **800+ tree plantation** campaign, **90% food-waste reduction**, a **1,000+ student pledge**, rural workshops and a 20+ artwork World Water Day initiative.

Business Development Team Lead, AIESEC

- **Organised** multiple competitions for **150+ village schoolchildren** (grades 1-6), broadening AIESEC's community engagement.
- Spearheaded a **societal-norms debate** for 100+ students and an environmental & **SDG awareness campaign** for 50+ university students, expanding AIESEC's social outreach.

EXTRACURRICULAR

- **Anchored** AWS community events for AWS UG Ahmedabad and Women in Tech; **co-organised** the **Prayogshala** booth demonstrating Amazon Q and PartyRock.
- Hospitality Manager for an international speaker at **TEDx CHARUSAT**.
- **Gold medalist**, football (women's team), Deista IIT Hyderabad.

HOBBIES AND INTERESTS

Reading novels, Painting, Cooking, Exploring new cuisines, Badminton, Watching anime, Vibe coding